MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-BVOC (HONOURS) Regular EXAMINATION April 2026, second semester

Skill Development Components (SDC) - MG2SDCFTA102 - FOOD ADDITIVES AND PACKAGING
(2025 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks:** 50

Part A

One Sentence

Answer any 10 questions
Each question carries 1 marks

1. List the classification of food additives based on intention. [K] / [CO1]

Ans - Direct food additives and Indirect food additives

2. Recall two control measures for preventing unintentional additives. [K] / [CO1]

Ans - GAP, GMP, Quality control, packaging control, proper storage condition, regulatory control, consumer awareness

3. Define E-numbering. [K] / [CO1]

Ans - E-numbers are codes given to food additives that are checked and approved for use in foods.

4. Define acidity regulators with an example. [K] / [CO1]

5. State the source of carotene used as a food colour. [K] / [CO2]

Ans - yellow colour, derived from carrot, is used in margarine

6. Define low-calorie sweeteners. [K] / [CO2]

Ans - 1–3 Kcal/gram, Eg - Sorbitol, Xylitol, Mannitol, Maltitol

7. Explain briefly how antioxidants prevent oxidation. [U] / [CO2]

Ans - Removing metal ions and stopping radical formation

8. Name any four humectants used in food products. [K] / [CO2]

Ans - propylene glycol, glycerol, and sorbitol and mannitol, xylitol

9. Define buffer solution. [K] / [CO2]

Ans - Buffer solutions can maintain a relatively constant pH when small amounts of acids or bases are added.

10. List the properties of TFS used in packaging. [K] / [CO3]

Ans - Good heat resistance, good lacquer adhesion, harder surface. Less corrosion resistant than tinplate

11. List two advantages of Controlled Atmosphere Packaging. [K] / [CO3]

Ans - Significantly increases shelf life, maintains freshness for long storage periods, Reduces post-harvest losses

12. Name any two chemical sterilizing agents used in aseptic packaging. [K] / [CO3]

Ans - Hydrogen peroxide (H_2O_2), Peracetic acid (PAA), Ethanol, Ethylene oxide and Various acids

Part B

Short Answer

Answer any 4 questions
Each question carries 5 marks

13. Define JECFA and CAC. [K] / [CO1]

JECFA

JECFA is an **independent scientific committee** created by the **Food and Agriculture Organization (FAO)** and the **World Health Organization (WHO)**. JECFA decides whether a food additive is safe and how much of it can be safely eaten.

Main functions:

- Evaluates the **safety of food additives**, contaminants, and residues.
- Conducts **risk assessments** based on scientific studies.
- Establishes **Acceptable Daily Intake (ADI)** values, which indicate how much of a substance can be safely consumed daily over a lifetime.

CAC

The Codex Alimentarius Commission was also established by **FAO and WHO**.

Main functions:

- Develops **international food standards**, guidelines, and codes of practice.
- Creates the **General Standard for Food Additives (GSFA)**, which lists:
    - Approved food additives

- Maximum permitted levels in different foods

- Helps protect **consumer health** and ensures **fair food trade** worldwide.

14. Define chemical leavening agents with suitable examples. [K] / [CO2]

Type	**Function**	**Common Uses**	**Action**
Baking Soda (Sodium Bicarbonate)	Releases CO_2 when reacts with acid (vinegar, lemon juice) and moisture	Cookies, cakes, muffins, pancakes	Baking soda + acid → CO_2 gas → dough rises
Baking Powder	Baking powder contains both an *acid* (usually cream of tartar - potassium hydrogen tartrate) and a *base* (baking soda), along with a *starch* to prevent premature reactions. When mixed with moisture, baking powder releases carbon dioxide gas.	Cakes, biscuits, muffins, quick breads	Reacts with moisture & heat → releases CO_2
Ammonium Bicarbonate (Baker's Ammonia)	Releases CO_2 and ammonia gas on heating	Crisp cookies, crackers	Gas released during heating → product rises

15. Differentiate between natural and synthetic curing agents with suitable examples. [U] / [CO2]

Natural Curing Agents: Derived from natural sources such as plants, minerals, spices, or naturally occurring food compounds. Eg - Salt, Natural Nitrates/Nitrites, Sugar, Spices

Synthetic Curing Agents: Chemically synthesized substances used in modern food processing for effective preservation and safety. Eg - Sodium Nitrite, Sodium Nitrate, Phosphates, Ascorbic Acid

16. Discuss the importance of sequestrants in maintaining quality and shelf life of processed foods. [U] / [CO2]

- It is also known as Sequestrants
- Chelating agents are substances that bind (complex) with transition metal ions like copper, iron, cobalt, and nickel (These metals are powerful catalysts in the auto-oxidation processes
- Purpose - Prevent oxidation by binding metal ions that act as catalysts in auto-oxidation.
- They need in food to prevent decolourisation, rancidity, and off-flavours in food products

17. Outline the main functions of each type of packaging system. [U] / [CO3]

Containment: Containment means keeping a product inside a container so that it can be safely stored and transported from the place where it is made to the place where it is used.

Preservation/Protection

Communication

Utility: This function of packaging is called convenience or utility.

18. Describe the method used to measure thickness using a micrometer. [U] / [CO3]

Thickness is the distance between the two surfaces of paper, paperboard, or flexible packaging material.

Method of Test: (Equipment name : micrometer)

- Cut a specimen of 10 cm × 10 cm.
- Place it between the two anvils of a micrometer.
- Gently lower the upper anvil and note the reading.
- Take at least 10 readings and calculate the average.
- Thickness is expressed in micron, inches, mil, or gauge.

Part C

Essay

Answer any 2 questions
Each question carries 10 marks

19. List out the properties of food additives. [K] / [CO1]

Preservation

- Preservation is one of the most important functions of food additives. Preservatives extend the shelf life of foods by inhibiting the growth of spoilage and microorganisms such as bacteria, yeasts and molds. Eg: potassium sorbate and sodium benzoate are

commonly used in beverages, sauces, pickles and bakery products to inhibit molds and yeasts.

- In addition to microbial control, oxidation (fat react with oxygen to form off-flavour amnd off-odour) is another major cause of food spoilage.

Antioxidants such as ascorbic acid (vitamin C) prevent oxidation of fats and oils.

Sensory Enhancement

- Taste enhancement: Flavouring agents and acids improve palatability (how tasty food is). Monosodium glutamate (MSG) is a flavour enhancer that intensifies umami taste in soups, sauces and processed foods
- Appearance (colour): Colour additives restore or enhance the natural colour lost during processing. Natural colours such as carotenoids, annatto, and chlorophyll.
- Texture modification: Emulsifiers like lecithin and mono- and diglycerides help in uniform mixing of oil and water in products such as mayonnaise and margarine. Thickeners and stabilizers such as pectin, guar gum and carrageenan improve viscosity, mouthfeel and consistency in sauces, jams and ice cream.
- Aroma enhancement: Flavouring substances contribute characteristic aromas, for example vanilla in bakery products or fruit esters in beverages. (smell)

Nutritional Enhancement

- Food additives play an important role in improving the nutritional quality of foods through fortification. Vitamins such as vitamin A, D and B-complex are added to milk, flour and cereals to prevent deficiency diseases. Minerals like iron, calcium and iodine are incorporated into salt, flour and beverages

Processing Aids

- Leavening agents, such as baking powder and baking soda, release carbon dioxide during baking, causing dough expansion and producing soft, porous bakery products.
- Anti-caking agents like silicon dioxide and calcium silicate prevent clumping in powdered foods such as salt, sugar and spice mixes.
- Enzymes used in food processing also fall under processing aids, improving efficiency and product quality. (ptns that speed up chemical reactions)

Safety and Stability

- Food additives must be safe for human consumption and stable under processing and storage conditions.
- Stability ensures that additives do not break down into harmful substances when exposed to heat, light or oxygen.

20. Define thickening agents in detail along with its types and functions in food. [U] / [CO2]

A thickening agent is something you add to a liquid to make it thicker without really changing its taste. In cooking, thickeners are often used to make sauces, soups, and puddings thicker and smoother. They work because their tiny molecules spread out in the liquid and get mixed together. This makes the liquid flow more slowly, so it feels thicker.

Types of Thickening Agents

1. Starch-Based Thickeners

Starches are the most common thickeners used in cooking. They come from plants and are mainly made of carbohydrates

Starch Type	Uses	Thickening Property	Special Characteristics
Cornstarch	Sauces, gravies, soups, puddings	Thickens when heated with water	Makes food smooth and slightly shiny; becomes gel-like when cooled
Arrowroot	Light sauces, baby food, fruit juices	Thickens at lower temperature than cornstarch	Forms clear gel; does not change taste or colour
Tapioca Starch	Pies, gravies, sauces, desserts	Strong thickening power	Gives smooth and shiny appearance; slightly chewy texture in desserts
Potato Starch	Soups, sauces, gravies, gluten-free cooking	Works at lower temperature	Gives smooth texture; not suitable for very acidic foods

2. Gelatin and Pectin

Gelatin and pectin are natural ingredients that turn liquids into a gel (a soft, jelly-like texture).

Gelatin
Gelatin is used to make foods like jelly, marshmallows, gummies, and some dairy desserts. It comes from animal collagen (usually from pork or beef). It dissolves in hot water and becomes firm and jelly-like when it cools.

Pectin (E440)
Pectin is mainly used in jams, jellies, and marmalades. It comes from fruits, especially apples

and citrus fruits. It forms a gel when mixed with sugar and acid, helping jams and jellies become thick and firm.

3. Gums (Hydrocolloids)

Gums are plant-based ingredients used to thicken liquids and keep ingredients mixed together. They are often used in sauces, salad dressings, dairy products, and gluten-free baking.

Thickening / Gelling Agent	Uses	Source	Special Properties
Xanthan Gum (E415)	Gluten-free baking, sauces, dressings, drinks	Produced by bacterial fermentation	Thickens liquids, prevents separation, retains moisture; effective in small amounts
Guar Gum (E412)	Dairy products, salad dressings, soups	Extracted from guar beans	Dissolves in water; works well in small quantities; good thickening ability
Agar-Agar (E406)	Jellies, ice creams, puddings	Obtained from red seaweed (algae)	Strong gelling agent; sets at higher temperature than gelatin
Carboxymethylcellulose (CMC) (E466)	Sauces, dressings, dairy products	Modified plant fiber (cellulose)	Dissolves easily in water; strong thickening ability

4. Modified Starches (E1400–E1452)

Starch is used in food mainly as a thickening and binding agent to improve texture, consistency, and stability of food products. Modified starches are starches that have been specially treated to work better in certain conditions, like high heat, acidic foods, or freezing and thawing. They are stronger and more stable than regular starch.

Instant or Pre-gelatinized Starch: This starch has already been cooked and dried. It can thicken food without heating. It is used in instant puddings, soups, and sauces that do not need cooking.

Cross-linked Starch: This type of starch is changed to make it stronger and more stable. It forms a stronger gel and can handle heat, acid, and freezing better than regular starch.

21. Classify preservatives into Class I and Class II along with their functions, examples, and permitted levels. [U] / [CO2]

A preservative is defined as any substance which is capable of inhibiting, retarding, or arresting, the growth of micro-organisms, of any deterioration of food due to micro-organisms, or of masking the evidence of any such deterioration.

Class I preservatives are natural substances and can be used without restriction e.g. salt, sugar, spices, vinegar.

Sodium chloride : Salts stop the growth of microorganisms and interfere with the action of proteolytic enzymes. Salts also cause food dehydration by drawing out water from the tissue cells. Used in butter, cheese, cabbage, olives, cucumbers, meat, fish and bread.

Sugar: Sugar acid in the preservation of products in which it is used. The high osmotic pressure of sugar creates conditions that are unfavourable for the growth and reproduction of most species of bacteria, yeasts and molds. Used in syrups and confectionary products, fondant fillings in chocolate, honey, jellies, marmalades, conserves and fruits such as dates, sultanas and currants.

Class II preservatives are man made and their use is restricted to only certain foods and the amount of the preservative which can be added to these foods is also specified under PFA rules.

Preservative	*E Number*	*Food Applications*	*Permitted Level**	*Main Function*
Sulfur dioxide	E220-E228	Treatment of fruits & vegetables before and after dehydration; storage of fresh grapes; wine making; manufacture of fruit juices; dried fruits; cut fruits	50–2000 mg/kg depending on food	Extends storage life; prevents growth of undesirable microorganisms and prevents browning reactions;
Sorbic acid	E200–E203	Cheese, baked products, fruit juices, wines, pickles	Up to 1000 mg/kg	Inhibits moulds and yeasts

Acetic acid	E260–E263	Pickled vegetables; bread and baked foods; mayonnaise and pickles	GMP	Prevents ropiness and mould growth
Propionic acid	E280–E283	Bakery products	Up to 2000–3000 mg/kg	Inhibits moulds; active against ropy bread organism *Bacillus mesentericus*
Benzoic acid	E210–E213	Fruit juices, carbonated beverages, pickles, sauerkraut	Up to 1000 mg/kg	Antimicrobial; most active against yeasts and bacteria
Parabens	E214–E219	General food preservation	Up to 300 mg/kg	Effective inhibitors of moulds and yeasts; relatively ineffective against bacteria.
Ethylene oxide & Propylene oxide	—-	Dry items such as nuts and spices	Not permitted directly in foods; only for fumigation of certain dry foods under strict regulation	Destroy all forms of microorganisms including spores and viruses

22. Explain the concept of intelligent packaging and describe the different types of indicators used in it. [U] / [CO3]

Intelligent packaging is defined as packaging that contains an external or internal indicator to provide information about aspects of the history of the package and/or the quality of the food.

External indicators are those indicators which are attached outside the package like time-temperature indicators

Internal indicators are those indicators which are placed inside the package, either to the head-space of the package or attached into the lid like oxygen indicators for indication of oxygen or package leak, carbon dioxide indicators, microbial growth indicators and pathogen indicators

Various examples of internal and external indicators are shown in the table below:

Time–Temperature Indicators (TTIs)

Time–Temperature Indicators record the temperature history of packaged foods during storage, transport, and sale. They provide information about food quality and safety through irreversible changes. TTIs are classified into diffusion, microbial, enzyme, polymerization, and light color types. They are mainly used for temperature-sensitive foods such as frozen and chilled foods.

Freshness Indicators

Freshness indicators show the freshness of food by detecting metabolites produced during spoilage. These indicators often produce visible color changes when reacting with compounds such as glucose, organic acids, ethanol, volatile nitrogen compounds, CO_2, sulfides, and biogenic amines. They are commonly used for fresh foods like fruits and seafood.

Leak Indicators

Leak indicators detect changes in gas composition inside food packages caused by respiration, microbial activity, or package leakage. They help monitor food quality and safety. Common gases detected include oxygen (O_2) and carbon dioxide (CO_2), while others like water vapor, ethanol, and hydrogen sulfide may also be monitored.

pH Indicators

pH indicators detect changes in pH levels of packaged foods. Natural compounds such as anthocyanins (polyphenols) are commonly used because they change color with pH variations, indicating food spoilage.